GMI German-Malaysian Institute

DIPLOMA PROGRAMME

PRACTICAL TEST 1

Academic Period: July 2025

Program	:	**NETWORK SECURITY**
Course	:	**SCALING NETWORK**
Code	:	**NTC 1093**
Duration	:	**2.5 HOURS**
Total Marks	:	**100**

GENERAL INSTRUCTIONS:

1. Write down clearly (with capital letters) your name, trade, group and the date of examination at the bottom of this page.
2. Do not open this question booklet until you are told to do so by the invigilator.
3. Read the specific instructions and questions carefully before answering.



This paper consists of **6** printed pages including this page.

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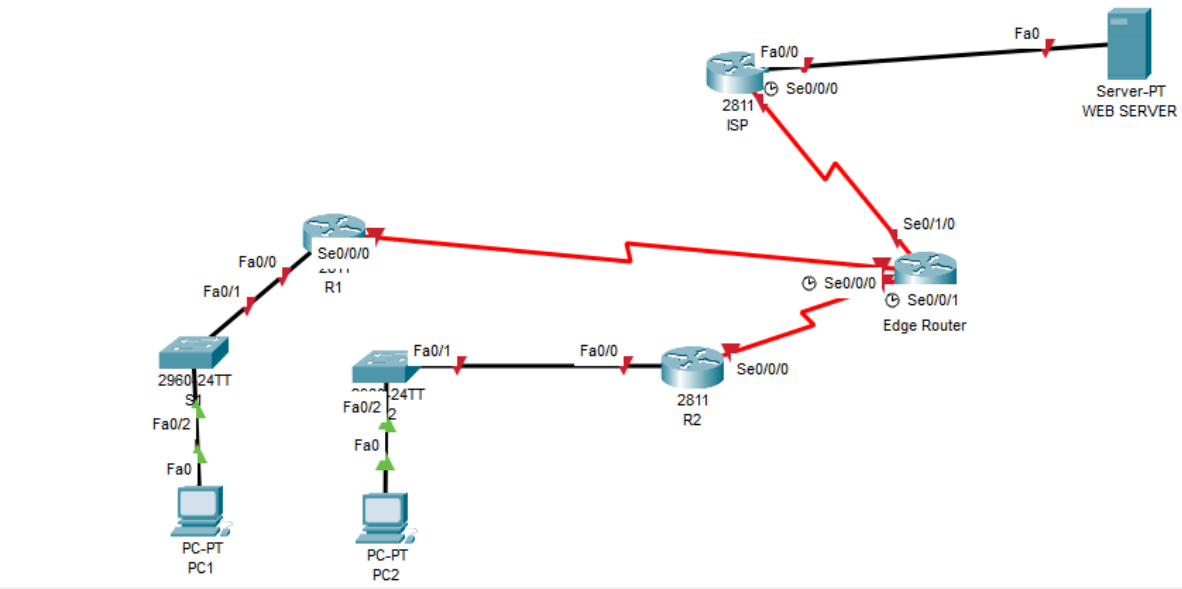
Semester/Trade/Group : **SEMESTER 3 / DCBS / DCBS 1**

Date : **30/9/2025**

No	Assessment Criteria	CLO	PO	Level of Diff	Full Marks	Score	Marks	Comment
1.	Setup Network	2	4	P3	3			
2	Configuration	2	4	P3	18			
3.	Verification	2	4	P3	3			
4.	Interpretation	2	4	P3	3			
5	Knowledge	2	4	P3	3			
				Total	30			
				Total Marks	/30			
		Total		**PO 2**	/30			
				Signature				

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NO.	CRITERIA	APPROACHES EXPECTATIONS [1]	MEETS EXPECTATIONS [2]	EXCELLENT [3]	MARKS
PRACTICAL TEST (1)					
1	**Setup Network** Cabling and setup network	Unable to setup network without major assistance.	Able to setup network with little assistance.	Able to setup network without assistance.	
2	**Configurations:** Solving the lab configuration for OSP and ACL	Unable to configure most of the OSPF and ACL (1-6)	Able to configure most of the OSPF and ACL (7-13)	Able to configure OSPF and ACL (14 - 18)	
3	**Verification:** Test connection	Unable to verify most of the configurations.	Students able to verify most of the configurations.	Student able to verify all configurations.	
4	**Interpretation:** Interpreting findings.	Unable to interpret most of the findings and result.	Able to interpret most of the findings and result.	Able to interpret all the findings and result clearly and comprehensively.	
5	**Knowledge**	Only restatement of the instruction without commenting on the expected key point. Incorrect judgment/arguments are used.	Shows a little understanding of what the practical was about.	Shows a full understanding of what the practical was about.	
	TOTAL				



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	S0/0/0	10.20.12.1	255.255.255.252	-
	F0/0	10.20.1.1	255.255.255.0	-
R2	S0/0/0	10.20.12.2	255.255.255.252	-
	F0/0	10.20.2.1	255.255.255.0	-
ISP	S0/0/0	203.0.113.2	255.255.255.252	-
	F0/0	198.51.100.1	255.255.255.0	-
Edge	S0/0/0	10.20.23.2	255.255.255.252	-
	S0/0/1	10.20.23.1	255.255.255.252	-
	S0/1/0	203.0.113.1	255.255.255.252	-
PC-A	NIC	10.20.1.10	255.255.255.0	10.20.1.1
PC-B	NIC	10.20.2.10	255.255.255.0	10.20.2.1
Web Server	NIC	198.51.100.100	255.255.255.0	198.51.100.1

Part 1: Build the Network and Configure Basic Device Settings

Step 1: Cable the network as shown in the topology.

Step 2: Configure PC hosts.

Step 3: Initialize and reload the routers and switches as necessary.

Step 4: Configure basic settings for each router.

Configuration tasks for router **R1** include the following:

Configuration Item or Task	Specification
1.Disable DNS lookup	`R1_Hariz(config)#no ip domain-lookup` `R1_Hariz(config)#`
2.Router name	`Router(config)#hostname R1_Hariz` `R1_Hariz(config)#`

3.IP addressing on each interface	`R1_Hariz(config)#int f0/0`
	`R1_Hariz(config-if)#ip address 10.20.1.1 255.255.255.0`
	`R1_Hariz(config-if)#no shut`
	`R1_Hariz(config)#int s0/0/0`
	`R1_Hariz(config-if)#ip address 10.20.12.1 255.255.255.252`
	`R1_Hariz(config-if)#no shut`

4.Save the configuration	`R1_Hariz(config)#end`
	`R1_Hariz#`
	`%SYS-5-CONFIG_I: Configured from console by console`
	`write memory`
	`Building configuration...`
	`[OK]`

5.Configure default route for R1, R2 and Edge	`R1_Hariz(config)#ip route 0.0.0.0 0.0.0.0 10.20.12.2` `R1_Hariz(config)#end`

6.Screen capture of all above tasks	`C:\>ping 10.20.2.10`
	
	`Pinging 10.20.2.10 with 32 bytes of data:`
	
	`Reply from 10.20.2.10: bytes=32 time=20ms TTL=125`
	`Reply from 10.20.2.10: bytes=32 time=13ms TTL=125`
	`Reply from 10.20.2.10: bytes=32 time=14ms TTL=125`
	`Reply from 10.20.2.10: bytes=32 time=33ms TTL=125`
	
	`Ping statistics for 10.20.2.10:`
	`Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),`
	`Approximate round trip times in milli-seconds:`
	`Minimum = 13ms, Maximum = 33ms, Average = 20ms`

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Configuration tasks for router **R2** include the following:

Configuration Item or Task	Specification
1.Disable DNS lookup	`R2_Hariz(config)#no ip domain-lookup` `R2_Hariz(config)#`
2.Router name	`R2(config)#hostname R2_Hariz` `R2_Hariz(config)#`

3.IP addressing on each interface	`R2_Hariz(config)#int f0/0`
	`R2_Hariz(config-if)#ip address 10.20.2.1 255.255.255.0`
	`R2_Hariz(config-if)#no shut`
	
	`R2_Hariz(config)#int s0/0/0`
	`R2_Hariz(config-if)#ip address 10.20.12.2 255.255.255.252`
	`R2_Hariz(config-if)#no shut`
	
	`R2_Hariz(config)#int s0/0/1`
	`R2_Hariz(config-if)#ip address 10.20.23.1 255.255.255.252`
	`R2_Hariz(config-if)#no shut`
	
	`%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down`
	`R2_Hariz(config-if)#exit`

4.Save the configuration	`R2_Hariz(config)#end`
	`R2_Hariz#`
	`%SYS-5-CONFIG_I: Configured from console by console`
	
	`R2_Hariz#write memory`
	`Building configuration...`
	`[OK]`
	`R2_Hariz#`

5. Screen capture of all above tasks	`C:\>ping 10.20.1.10`
	
	`Pinging 10.20.1.10 with 32 bytes of data:`
	
	`Reply from 10.20.1.10: bytes=32 time=2ms TTL=125`
	`Reply from 10.20.1.10: bytes=32 time=15ms TTL=125`
	`Reply from 10.20.1.10: bytes=32 time=16ms TTL=125`
	`Reply from 10.20.1.10: bytes=32 time=12ms TTL=125`
	
	`Ping statistics for 10.20.1.10:`
	`Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),`
	`Approximate round trip times in milli-seconds:`
	`Minimum = 2ms, Maximum = 16ms, Average = 11ms`

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Configuration tasks for router **EDGE** include the following:

Configuration Item or Task	Specification
1.Disable DNS lookup	EDGE_Hariz(config)#no ip domain-lookup EDGE_Hariz(config)#
2.Router name	Router(config)#hostname EDGE_Hariz EDGE_Hariz(config)#
3.IP addressing on each interface	:DGE_Hariz(config)#int s0/0/0 :DGE_Hariz(config-if)#ip address 10.20.23.2 255.255.255.252 :DGE_Hariz(config-if)#no shut :DGE_Hariz(config-if)#exit EDGE_Hariz(config)#int s0/1/0 EDGE_Hariz(config-if)#ip address 203.0.113.1 255.255.255.252 EDGE_Hariz(config-if)#no shut EDGE_Hariz(config-if)#exit
4.Save the configuration	EDGE_Hariz(config)#end EDGE_Hariz# %SYS-5-CONFIG_I: Configured from console by console write memory Building configuration... [OK] EDGE_Hariz#
5. Screen capture of all above tasks	PC1 Minimum = 13ms, Maximum = 33ms, Average = 20ms C:\>ping 198.51.100.100 Pinging 198.51.100.100 with 32 bytes of data: Reply from 198.51.100.100: bytes=32 time=8ms TTL=125 Reply from 198.51.100.100: bytes=32 time=5ms TTL=125 PC2 C:\>ping 198.51.100.100 Pinging 198.51.100.100 with 32 bytes of data: Reply from 198.51.100.100: bytes=32 time=24ms TTL=125 Reply from 198.51.100.100: bytes=32 time=19ms TTL=125 Reply from 198.51.100.100: bytes=32 time=12ms TTL=125 Reply from 198.51.100.100: bytes=32 time=18ms TTL=125

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Part 2: Configure and verify OSPF and ACL

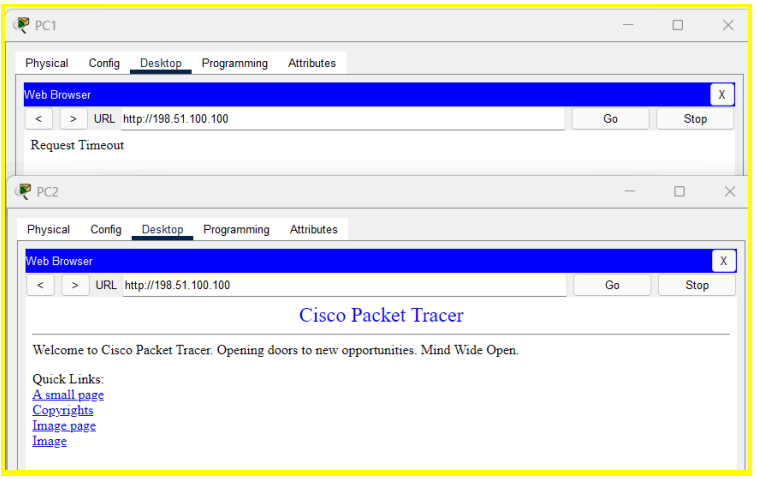
Step 1: Configure OSPF and ACL on R1, R2 and EDGE.

Configuration Item or Task	Specification

1.Configure OSPF to ensure all routers are accessible	R1 :
	`R1_Hariz(config)#router ospf 1`
	`R1_Hariz(config-router)#network 10.20.1.0 0.0.0.255 area 0`
	`R1_Hariz(config-router)#network 10.20.12.0 0.0.0.3 area 0`
	R2 :
	`R2_Hariz(config)#router ospf 1`
	`R2_Hariz(config-router)#network 10.20.2.0 0.0.0.255 area 0`
	`R2_Hariz(config-router)#network 10.20.12.0 0.0.0.3 area 0`
	`R2_Hariz(config-router)#network 10.20.23.0 0.0.0.3 area 0`
	Edge :
	`EDGE_Hariz(config)#router ospf 1`
	`EDGE_Hariz(config-router)#network 10.20.23.0 0.0.0.3 area 0`

2.Show IP route for each router	R1 :
	`Gateway of last resort is not set`
	`10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks`
	`C 10.20.1.0/24 is directly connected, FastEthernet0/0`
	`L 10.20.1.1/32 is directly connected, FastEthernet0/0`
	`S 10.20.2.0/24 [1/0] via 10.20.13.1`
	`C 10.20.13.0/30 is directly connected, Serial0/0/0`
	`L 10.20.13.2/32 is directly connected, Serial0/0/0`
	`S 198.51.100.0/24 [1/0] via 10.20.13.1`
	R2 :
	`Gateway of last resort is 10.20.12.1 to network 0.0.0.0`
	`10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks`
	`C 10.20.2.0/24 is directly connected, FastEthernet0/0`
	`L 10.20.2.1/32 is directly connected, FastEthernet0/0`
	`C 10.20.12.0/30 is directly connected, Serial0/0/0`
	`L 10.20.12.2/32 is directly connected, Serial0/0/0`
	`S* 0.0.0.0/0 [1/0] via 10.20.12.1`
	Edge :
	`Gateway of last resort is 203.0.113.2 to network 0.0.0.0`
	`10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks`
	`S 10.20.1.0/24 [1/0] via 10.20.13.2`
	`S 10.20.2.0/24 [1/0] via 10.20.23.2`
	`C 10.20.13.0/30 is directly connected, Serial0/0/0`
	`L 10.20.13.1/32 is directly connected, Serial0/0/0`
	`C 10.20.23.0/30 is directly connected, Serial0/0/1`
	`L 10.20.23.1/32 is directly connected, Serial0/0/1`
	`S 198.51.100.0/24 [1/0] via 203.0.113.2`
	`203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks`
	`C 203.0.113.0/30 is directly connected, Serial0/1/0`
	`L 203.0.113.1/32 is directly connected, Serial0/1/0`
	`S* 0.0.0.0/0 [1/0] via 203.0.113.2`

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3. Apply ACL of the following a. PC1 cannot browse to Web server b. PC2 cannot use Telnet to Web server c. Allow all other traffic	EDGE_Hariz#conf t Enter configuration commands, one per line. End with CNTL/Z. EDGE_Hariz(config)#access-list 100 deny tcp host 10.20.1.10 host 198.51.100.100 eq 80 EDGE_Hariz(config)#access-list 100 deny tcp host 10.20.2.10 host 198.51.100.100 eq 23 EDGE_Hariz(config)#access-list 100 permit ip any any EDGE_Hariz(config)#access-list 100 permit tcp host 10.20.2.10 host 198.51.100.100 eq 80 EDGE_Hariz(config)#access-list 100 permit tcp host 10.20.1.10 host 198.51.100.100 eq 23 EDGE_Hariz(config)#end
.4. Apply to correct interface	EDGE_Hariz(config)#int s0/1/0 EDGE_Hariz(config-if)#ip access-group 100 out EDGE_Hariz(config-if)#exit
5. Screen capture of all above tasks	

Step 2:

Configuration Item or Task	Specification
1.Show ACL here (Ensure ACL match)	EDGE_Hariz# %SYS-5-CONFIG_I: Configured from console by console show access-lists Extended IP access list 100 10 deny tcp host 10.20.1.10 host 198.51.100.100 eq www 20 deny tcp host 10.20.2.10 host 198.51.100.100 eq telnet 30 permit ip any any 40 permit tcp host 10.20.2.10 host 198.51.100.100 eq www 50 permit tcp host 10.20.1.10 host 198.51.100.100 eq telnet

Submission in VLU:

1) Packet Tracer
2) Lab sheet (in **pdf format**)

Format should be CBS1_yourname.pdf
Format should be CBS1_yourname.pkt

-THE END-

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